

Solutions to Quiz 11

Question 1

In each of the following cases decide whether $R = \mathbb{Z}[\alpha]$ is the ring $\mathcal{O}_K$ of all algebraic integers in $K = \mathbb{Q}[\alpha]$.

1. $\alpha = \sqrt{-7}$.
2. α satisfies $X^3 - 3X + 1$.

Solution:

1. We note that $\beta = (1 + \sqrt{-7})/2$ satisfies the equation $X^2 - X + 2 = 0$. Hence, it is an algebraic integer in $\mathbb{Q}[\sqrt{-7}]$. Since β not of the form $p + q\sqrt{-7}$ for p, q integers we see that β is not in $\mathbb{Z}[\alpha]$. It follows that $R \neq \mathcal{O}_K$ in this case.
2. We note that the discriminant of $X^3 - 3X + 1$ is $-4 \cdot (-3)^3 - 27 \cdot (1)^2 = 81$. It follows that $\mathbb{Z}_p \otimes_{\mathbb{Z}} R = \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathcal{O}_K$ for $p \neq 3$. In other words, we only need to check this equality at $p = 3$.

For $p = 3$, we see that $P(X)$ is congruent to $(X + 1)^3$ modulo 3. Thus, we substitute $Y = X + 1$ (equivalently $X = Y - 1$ to get

$$(Y - 1)^3 - 3(Y - 1) + 1 = Y^3 - 3Y^2 + 3$$

which is an Eisenstein polynomial in $\mathbb{Z}_3[Y]$. It follows that $\mathbb{Z}_3[Y]/\langle Y^3 - 3Y^2 + 3 \rangle$ consists of all algebraic integers in $\mathbb{Q}_3[Y]/\langle Y^3 - 3Y^2 + 3 \rangle$. By reversing the change of co-ordinates back to X , we see that the same is true for $\mathbb{Z}_3[\alpha]$.